

Wilson Wu

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Education

University of California, Los Angeles

B.S. in Electrical Engineering | GPA: 3.59

Los Angeles, CA

Expected June 2028

- **Honors:** ECE Fast Track Honors Program (Top 20)
- **Relevant Coursework:** Circuit Theory, Signals and Systems, Electricity and Magnetism, Internet of Things, Object-Oriented Programming (C++), Data Structures and Algorithms

Experience

Bruin Supermileage — Hydrogen Fuel-Cell Vehicle

June 2026 – Present

Powertrain Project Engineer | KiCad, PSpice, LTspice

Los Angeles, CA

- Developing a **brake-chopper PCB** targeting 3.6 kW of regenerative-braking power dissipation through an external resistor to keep the vehicle's DC bus below the 60 V competition limit
- Comparing hysteresis bands, MOSFET-driver configurations, and braking resistors in **LTspice** and **PSpice** to evaluate switching response, bus-voltage regulation, and power dissipation

Low Voltage Engineer | KiCad, LTspice, CAN, Embedded Systems

October 2025 – June 2026

- Designed a **two-layer safety PCB** integrating two hydrogen sensors, an ignition interlock, emergency-stop inputs, and CAN communication for a Shell Eco-marathon vehicle
- Developed a **hardware interlock** combining hydrogen-sensor, ignition, and emergency-stop signals, reducing safety-board response time by 1 s compared with the previous year's design
- Verified **CAN communication and relay response** through bench and in-vehicle testing across four fault conditions

UCLA Plasma Accelerator/Laser-Plasma Interactions Group

June 2026 – Present

Undergraduate Researcher | Python, OSIRIS, HDF5

Los Angeles, CA

- Reproduced published nonlinear plasma-wave amplitude-growth results by implementing **numerical models** of autoresonant beat-wave excitation
- Configured **OSIRIS particle-in-cell simulations**, tuning grid resolution and particle count to reproduce plasma-wave amplitudes with 80% accuracy relative to numerical-model predictions
- Optimized a **Python pipeline** to process 337 MB of HDF5 simulation output, reducing processing time by 75% (2 min to 30 s) and using FFT and Hilbert transforms to extract spectra and amplitude envelopes

Projects

Micromouse: Autonomous Maze-Solving Robot | KiCad, STM32, C

December 2025 – Present

- Led a **five-person team** developing a maze-solving robot, coordinating hardware and firmware work across power, sensing, and motor-control subsystems
- Designed a **two-layer mixed-signal PCB** with four infrared emitter-detector pairs and a gyroscope, separating sensor traces from high-current motor-driver routing to limit interference
- Restored functionality to a fabricated PCB by tracing layout and STM32 firmware errors, then applying jumper-wire and code fixes
- Developed **flood-fill logic** in STM32 C to map visited maze cells and update path costs

Autonomous Line-Following Car | C++, PD Control, Sensors

Spring 2026

- Tuned a **proportional-derivative controller** across multiple vehicle speeds, reducing tracking error by 95%
- Diagnosed sensor-alignment bias from **serial data** and added calibration offsets and turn-detection logic to improve tracking stability

Gacha Thrift: Clothing Dispenser — IdealHacks 2026 | ESP32-S3, C++, Python

February 2026

- Built a working thrift-clothing dispenser in **36 hours** by integrating the mechanical assembly, ESP32-S3 controls, and browser interface
- Synchronized physical dispensing with browser animations using **encoder-feedback motor control** on the ESP32-S3 and a Python WebSocket bridge

Technical Skills

Programming: Python, C/C++, Java, Git, Docker

Scientific Computing: NumPy, SciPy, Matplotlib, HDF5, OSIRIS (particle-in-cell simulation)

Hardware and Embedded: KiCad, LTspice, PSpice, STM32, ESP32-S3, CAN

Lab: Oscilloscope, DMM, Bench Power Supply, Soldering